

A Residual-Based Quantum Linear System Algorithm with Dynamic Stopping and Applications to Elliptic PDEs

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Abstract

Quantum linear-system algorithms (QLSAs) have rigorous worst-case complexity guarantees, but their runtimes are often chosen from spectral information assumed in advance. What is largely lacking is an a posteriori progress flag: most QLSA workflows, unlike the classical counterparts, do not provide a built-in mechanism to signal whether a particular instance has already converged.

For discretizations of elliptic PDEs $-\nabla \cdot (a(x)\nabla u(x)) = f(x)$, with divergence-gradient structure

$$-\nabla \cdot (a(x)\nabla) \approx A_h = G_h^\dagger G_h,$$

we formulate a stable first-order ODE whose limiting solution block is the desired Galerkin solution. The PDE-dependent scale is then $\|G_h\| = \mathcal{O}(h^{-1})$, comparable to factorized QLSA constructions with square-root condition-number scaling. We design an augmented dynamics with residual variables, in which measuring a residual register gives an on-the-fly convergence indicator without reconstructing the solution vector. For smooth right-hand sides, dynamic stopping can reduce the evolution time and gate count relative to a fixed worst-case schedule, and may also reduce exposure to accumulated hardware errors. Numerical experiments for a two-dimensional finite element Poisson problem show that the residual-register probability follows the actual error decay and, for some right-hand sides, can stop the quantum circuit well before a conservative worst-case runtime estimate is reached.

1 Introduction

Quantum linear-system algorithms (QLSAs) have become one of the central primitives in quantum algorithms. Since the HHL algorithm [20], substantial progress has been made in improving the dependence on precision, condition number, and access models, including variable-time amplitude amplification [1], exponentially improved precision dependence [12], block encodings and quantum singular value transformation [18], and adiabatic or adiabatic-inspired linear-system solvers [36, 4, 16]. At the same time, QLSAs are not merely standalone algorithms. They are used as subroutines in quantum algorithms for differential equations, least-squares problems, and other scientific-computing tasks. Therefore, the practical behavior of a QLSA can directly affect the performance of a larger quantum algorithm.

Most QLSA guarantees fix the algorithmic effort in advance, for example, a polynomial degree, an adiabatic schedule length, or a simulation time, from worst-case spectral information such as a condition-number bound or a promised singular-value interval. While this is the right framework for proving black-box complexity theorems, it is different from the way linear systems are solved in

practice: Classical iterative solvers typically do not run only for a fixed worst-case iteration count. Instead, they monitor residuals and stop when the current instance has converged. In this paper, we ask whether a comparable residual-based stopping mechanism can be incorporated into a QLSA.

This question is particularly natural for elliptic partial differential equations (PDEs), which are a standard source of large sparse linear systems in scientific computing. After discretization, a second-order elliptic operator usually gives a matrix A_h with condition number

$$\kappa(A_h) = \Theta(h^{-2}), \quad (1)$$

where h represents the scale of the mesh size. This worst-case scale is determined by the highest-frequency modes supported by the mesh. A particular right-hand side f , however, may be smooth and thus have little spectral weight in those modes. Consequently, the practical cost of solving the instance may depend not only on $\kappa(A_h)$, but also on the spectral content of the discrete vector $\mathbf{b}_h$. This type of right-hand-side dependence has begun to appear explicitly in the QLSA literature: for example, Li's instance-aware QLSA uses parameters of the augmented system $H = [A, -\mathbf{b}]$, together with a structure-aware right-rescaling, to obtain bounds that depend on the alignment of $\mathbf{b}$ with the linear system rather than only on $\kappa(A)$ [29]. Our approach is complementary. We do not replace the worst-case complexity theory. Rather, we formulate a QLSA as a time evolution and introduce a residual variable into the dynamics so that convergence can be detected during the computation.

The elliptic problems considered here have a discrete divergence–gradient factorization

$$A_h = G_h^\dagger G_h,$$

that comes from Galerkin projections in general. For finite differences and fixed-order finite elements, the first-order factor G_h is sparse and has operator norm

$$\|G_h\| = \Theta(h^{-1}) = \Theta\left(\sqrt{\kappa(A_h)}\right), \quad (2)$$

whereas the stiffness matrix itself has scale $\|A_h\| = \Theta(h^{-2})$. This factorization can be leveraged for QLSAs. In particular, Orsucci and Dunjko studied factorized positive-definite systems $A = LL^\dagger$ and showed how the factor L can be used to obtain square-root condition-number scaling [33]. In the elliptic setting, L naturally emerges from the discrete divergence–gradient factor G_h .

Our first observation is that the same first-order scaling can be obtained from a stable ordinary differential equation (ODE) formulation: the solution of the linear system appears as the limiting solution block of the dynamics and the simulation complexity scales as $\mathcal{O}(h^{-1})$. It is worthwhile to mention that this dynamical viewpoint is familiar in classical numerical analysis. Richardson iteration and related indirect methods for elliptic PDEs can be interpreted as time discretizations of parabolic PDE dynamics whose steady state is the desired elliptic solution [39, Chapter 8]. The distinct novelty here is that our ODE formulation also carries residual variables. These variables are not part of the solution vector, but they are essential for deciding whether the computation has converged. In classical numerical linear algebra [19, 35], the residual

$$\mathbf{r}_h = \mathbf{b}_h - A_h \mathbf{x}_h$$

is a basic diagnostic. It is tied to the algebraic error

$$\mathbf{e}_h = \mathbf{x}_* - \mathbf{x}_h$$

through

$$A_h \mathbf{e}_h = \mathbf{r}_h.$$

Our goal is to build an analogue of this mechanism into a quantum-compatible dynamical system. We formulate two related results. The first result shows that the elliptic problem can be written as a stable first-order ODE whose quantum simulation scale is $\mathcal{O}(\sqrt{\kappa})$. The second result shows that the added residual register can be measured to enable dynamic stopping.

Theorem 1.1 (ODE-based QLSA). *Let $A_h = G_h^\dagger G_h$ be the normalized stiffness matrix for the elliptic discretization. Introduce dynamic variables,*

$$\mathbf{w} = \begin{bmatrix} \mathbf{r} \\ \mathbf{s} \end{bmatrix}, \quad M_h = \begin{bmatrix} 0 & -G_h^\dagger \\ G_h & -I \end{bmatrix}, \quad P_r = \begin{bmatrix} I & 0 \end{bmatrix}, \quad (3)$$

and consider the accumulator relaxation

$$\dot{\mathbf{x}} = P_r \mathbf{w}, \quad \dot{\mathbf{w}} = M_h \mathbf{w}, \quad \mathbf{x}(0) = 0, \quad \mathbf{w}(0) = \begin{bmatrix} \mathbf{b}_h \\ 0 \end{bmatrix}. \quad (4)$$

Then $\mathbf{w}(t)$ decays exponentially and $\mathbf{x}(t)$ converges to the Galerkin solution $\mathbf{x}_*$ satisfying

$$G_h^\dagger G_h \mathbf{x}_* = \mathbf{b}_h.$$

Suppose that G_h/α_G admits sparse or block-encoding access with

$$\alpha_G = \Theta(\|G_h\|),$$

and that the right-hand-side state $|\mathbf{b}_h\rangle$ can be prepared with cost C_b . Then, for any target accuracy ε , a stable quantum linear-ODE primitive with linear dependence on $\alpha_G T$ prepares an ε -accurate normalized solution state with leading query and gate complexity

$$\tilde{\mathcal{O}}\left(C_b + \alpha_G \text{polylog} \frac{1}{\varepsilon}\right),$$

up to the standard output-normalization factor. Moreover, after the relaxation time $T = \tilde{\mathcal{O}}(\log(1/\varepsilon))$, the $\mathbf{x}$ -block has order-one weight in the joint state, $(\mathbf{w}, \mathbf{x})$, so extracting the solution block has constant overhead.

For elliptic discretizations with $\lambda_{\min}(A_h) = \Theta(1)$, $\lambda_{\max}(A_h) = \Theta(h^{-2})$, $\|G_h\| = \Theta(h^{-1})$, this gives

$$\alpha_G = \Theta(h^{-1}) = \Theta\left(\sqrt{\kappa(A_h)}\right),$$

and hence the mesh-dependent complexity scaling is

$$\tilde{\mathcal{O}}\left(\sqrt{\kappa(A_h)} \text{polylog} \frac{1}{\varepsilon}\right).$$

To run the dynamics in Eq. (4), suitable implementation primitives include QLSA-based ODE solvers, LCHS, Schrödingerisation, Laplace-transform eigenvalue-transformation methods, and recent algorithms for stable linear nonunitary dynamics [6, 28, 5, 2, 26, 3, 31, 30, 22].

The estimate in Theorem 1.1 is comparable to factorized QLSA constructions based on square-root condition-number scaling [33]. It is still, however, a worst-case statement. It does not tell us whether a particular right-hand side requires the full worst-case polynomial degree, ODE time, or simulation effort. To illustrate this point, we consider the standard Poisson problem

$$-\Delta u = f \quad \text{in } \Omega = (0, 1)^2, \quad u = 0 \quad \text{on } \partial\Omega,$$

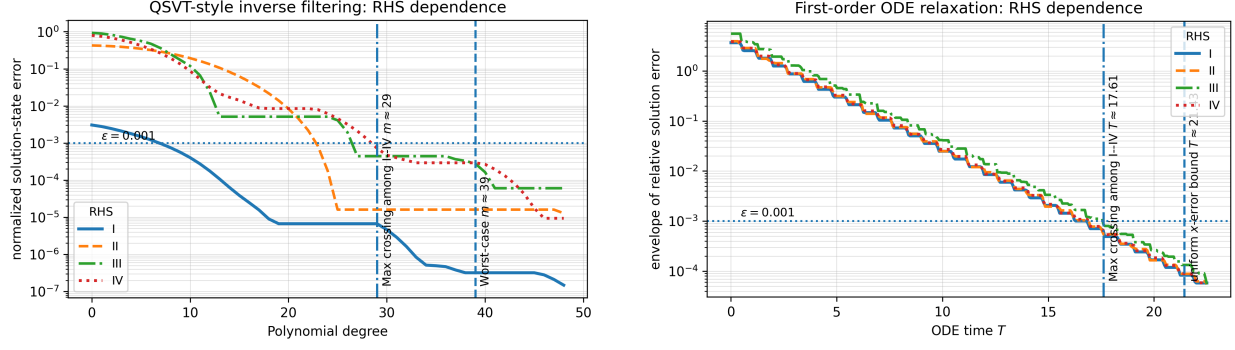


Figure 1: Right-hand-side dependence for the finite element Poisson problem. Left: a QSVT-style spectral inverse-polynomial experiment, where the horizontal axis is the polynomial degree. This is a spectral-polynomial emulator for the inverse filter rather than a hardware-level QSVT circuit. Right: the first-order ODE relaxation Eq. (4), where the horizontal axis is the ODE evolution time. The ODE curves use the relative solution error. In both panels, the dashed vertical line is the conservative worst-case prediction, while the dash-dotted line marks the latest empirical crossing among the benchmark cases I–IV.

discretized by continuous piecewise-linear finite elements on a uniform triangular mesh obtained by splitting each square cell into two triangles. We test QLSAs for four choices of the right-hand side functions (detailed setup in Section 5):

$$\begin{aligned}
 \text{I: } f_{\text{I}}(x, y) &= 2\pi^2 \sin(\pi x) \sin(\pi y), \\
 \text{II: } f_{\text{II}}(x, y) &= 2\pi^2 \sin(\pi x) \sin(\pi y) + \frac{5\pi^2}{2} \sin(2\pi x) \sin(\pi y), \\
 \text{III: } f_{\text{III}}(x, y) &= 2\pi^2 \sin(\pi x) \sin(\pi y) + \frac{13\pi^2}{2} \sin(3\pi x) \sin(2\pi y) + \frac{41\pi^2}{4} \sin(5\pi x) \sin(4\pi y), \\
 \text{IV: } f_{\text{IV}}(x, y) &= \exp\left(-\frac{(x - 1/2)^2 + (y - 1/2)^2}{0.1^2}\right).
 \end{aligned}$$

Cases I–III are increasingly oscillatory sine combinations, while Case IV is a localized smooth bump. The matrix A_h and the first-order factor G_h are fixed by the mesh; only the right-hand side changes. These are standard manufactured-solution tests for elliptic discretizations [8, 14].

In Fig. 1, the left panel implements a QSVT-style inverse-polynomial experiment. Here QSVT refers to quantum singular value transformation [18]: given a block encoding of a matrix, QSVT implements a polynomial transformation with a spectral inverse-polynomial filter. We plot the normalized solution-state error as a function of the polynomial degree. The right panel shows the first-order ODE relaxation Eq. (4) starting with zero initial condition. One observes that the practical degree or evolution time can depend strongly on the right-hand side. Some input functions lead to solutions that cross the target tolerance $\epsilon = 10^{-3}$ well before the conservative worst-case line. Therefore, a fixed runtime can be overly conservative: the algorithm is guaranteed to succeed, but the circuit might be deep and the particular instance may have converged much earlier.

The ODE formulation in Eq. (4) also has another advantage that is not native to the standard one-shot QSVT inverse-filter workflow. It can naturally incorporate an initial guess. A warm start can also be incorporated by choosing an initial $\mathbf{x}(0)$ and an auxiliary $\mathbf{q}(0)$, then setting

$$\mathbf{r}(0) = \mathbf{b}_h - G_h^\dagger \mathbf{q}(0), \quad \mathbf{s}(0) = G_h \mathbf{x}(0) - \mathbf{q}(0).$$

If these residuals are small or concentrated in rapidly damped modes, the observed relaxation time can be reduced. The ODE formulation also makes it natural to organize residual readout through checkpoint circuits. Rather than estimating the solution error directly, we monitor residual information carried by the dynamics. The next result states the corresponding residual-augmented version of the first-order solver.

Theorem 1.2 (ODE-based QLSA with dynamic stopping (Informal)). *Under the hypotheses of Theorem 1.1, and assume that the joint dynamics carrying the solution block $\mathbf{x}(t)$ and a mixed residual block is implemented by a quantum ODE solver.*

Let $t_{\text{ent}}(\mathbf{b}_h)$ be the first checkpoint time at which a constant-threshold residual test accepts, meaning that the measured residual probability indicates entry into the exponentially decaying residual regime. Let

$$T_{\star}(\mathbf{b}_h, \varepsilon) = t_{\text{ent}}(\mathbf{b}_h) + \Delta t(\varepsilon)$$

be the final production time, where $\Delta t(\varepsilon) = \mathcal{O}(\log(1/\varepsilon))$ is the deterministic coasting time chosen from the stability estimate. Then a production run at time $T_{\star}$ prepares an ε -accurate normalized solution state with leading cost

$$\tilde{\mathcal{O}}\left(C_b + \alpha_G T_{\star}(\mathbf{b}_h, \varepsilon) \text{polylog} \frac{1}{\varepsilon}\right),$$

up to the standard output-normalization factors of the chosen nonunitary ODE primitive.

We will show that the checkpoint test uses a constant residual-probability threshold, independent of ε . Thus the stopping test does not require resolving an $\mathcal{O}(\varepsilon^2)$ -probability event by direct sampling. The corresponding threshold and sampling model are elaborated in Sections 4.2 to 4.4.

Overall, the residual-augmented accumulator ODE keeps the solution of the linear system as its limiting $\mathbf{x}$ -block, while adding a measurable indicator of progress. Thus the purpose of the residual dynamics is not to improve the optimal worst-case black-box QLSA complexity. Rather, it is to make the solver more adaptive: run the dynamics, monitor a residual register, detect when the current instance has entered a small-residual regime, and then use the stability estimate to finish the solve with a prescribed accuracy.

Related work. Quantum linear-system algorithms have developed along several complementary directions since HHL [20]. Improvements based on variable-time amplitude amplification, precision-efficient linear-system solvers, block encodings, and QSVT have led to nearly optimal black-box QLSA complexity under standard oracle assumptions [1, 12, 18]. A separate line of work develops adiabatic and adiabatic-inspired linear-system solvers, in which the solution state is prepared by following, randomizing, or discretizing a Hamiltonian path associated with the linear system [36, 4, 16, 23]. These algorithms provide a conceptually different route to QLSA with near-optimal or optimal dependence on the condition number and precision, but their stopping times are still prescribed from a priori spectral information.

Quantum algorithms for elliptic PDEs have also been studied through finite difference, finite element, and spectral discretizations [11, 15, 32, 13]. More recent work has emphasized explicit block encodings, boundary conditions, and quantum-compatible preconditioning [17, 27]. Preconditioning is particularly important for PDE linear systems, and multilevel methods such as the BPX preconditioner provide a natural bridge between classical elliptic theory and quantum linear algebra [7, 17]. Tong, An, Wiebe, and Lin developed fast inversion as a preconditioning primitive and discussed QLSP costs in terms of right-hand-side overlap and effective condition

numbers [38]. Schrödingerization-based approaches have also been used to connect classical iterative linear-algebra methods and preconditioned linear systems to quantum simulation frameworks [24, 25]. Recent quantum multigrid work for finite element problems also seeks to bring multi-level iterative-solver structure into the quantum setting, particularly to address condition-number growth and the use of initial guesses [34]. This is complementary to the present work: qMG introduces a quantum analogue of multigrid iteration, whereas our focus is on a co-evolving residual register and a residual-based stopping test. These works are closely related in spirit because they bring numerical linear-algebra structure into quantum algorithms. The present paper differs in that the residual is embedded into the quantum dynamics as a measurable register.

Another relevant direction is instance-aware QLSA. Standard complexity estimates are often stated in terms of matrix-level parameters such as sparsity, normalization, and condition number. In practice, however, the difficulty of a linear-system instance can depend strongly on the input state or right-hand side. Li’s right-hand-side dependent QLSA [29] is a recent example of this perspective.

Organization. The remainder of the paper develops this idea in the elliptic setting. We first review the Galerkin discretization and the divergence–gradient factorization of the stiffness matrix in Section 2. We then explain sparse and block-encoding access to the first-order factor G_h . Next in Section 3 we formulate the first-order ODE solver and derive its stability and complexity estimates. We then augment these ODEs with residual variables, dilate the resulting nonunitary dynamics to a Schrödinger equation, and show how measuring a residual register yields a residual-based dynamic stopping rule in Section 4. Finally, we present numerical experiments in Section 5 and discuss extensions in Section 6.

2 Elliptic discretizations and first-order access

In this section we review the finite element linear systems that will be used throughout the paper. All objects in this section are classical matrices and vectors. We reserve bra-ket notation, such as $|\mathbf{b}_h\rangle$ and $|\mathbf{x}_h\rangle$, for the next section, where these vectors are encoded as normalized quantum states.

Let $\Omega \subset \mathbb{R}^d$ be a bounded connected Lipschitz domain and consider the elliptic model problem

$$-\nabla \cdot (a(x)\nabla u(x)) = f(x), \quad x \in \Omega, \quad u|_{\partial\Omega} = 0. \quad (5)$$

We assume that $a(x) \in \mathbb{R}^{d \times d}$ is symmetric and uniformly elliptic:

$$a_{\min}|\boldsymbol{\psi}|^2 \leq \boldsymbol{\psi}^T a(x) \boldsymbol{\psi} \leq a_{\max}|\boldsymbol{\psi}|^2 \quad \text{for all } \boldsymbol{\psi} \in \mathbb{R}^d.$$

The weak formulation is: find $u(x) \in H_0^1(\Omega)$ such that

$$\mathbf{a}(u, v) = \mathbf{f}(v) \quad \text{for all } v \in H_0^1(\Omega), \quad (6)$$

where from integration by parts, the bilinear form and linear functional are respectively given by,

$$\mathbf{a}(u, v) = \int_{\Omega} \nabla v(x)^T a(x) \nabla u(x) dx, \quad \mathbf{f}(v) = \int_{\Omega} f(x) v(x) dx.$$

This variational form is the starting point of the Galerkin finite element method [14, 8]. Instead of enforcing the PDE pointwise, one asks that the equation hold after testing against all functions in the energy space.

2.1 Galerkin discretization

Let $\mathcal{T}_h$ be a shape-regular, quasi-uniform triangulation with mesh width h , and let $V_h \subset H_0^1(\Omega)$ be a conforming finite element space with basis $\{\phi_j\}_{j=1}^{N_h}$. The Galerkin approximation seeks a solution in the subspace spanned by these basis functions,

$$u_h(x) = \sum_{j=1}^{N_h} (\mathbf{x}_h)_j \phi_j(x) \in V_h$$

and is determined by forcing the weak form (6) in the subspace,

$$\mathbf{a}(u_h, v_h) = \mathbf{f}(v_h) \quad \text{for all } v_h \in V_h. \quad (7)$$

Taking $v_h = \phi_k$ gives the finite-dimensional linear system

$$A_h \mathbf{x}_h = \mathbf{b}_h, \quad (8)$$

where

$$(A_h)_{kj} = \int_{\Omega} \nabla \phi_k(x)^T a(x) \nabla \phi_j(x) dx, \quad (\mathbf{b}_h)_k = \int_{\Omega} f(x) \phi_k(x) dx. \quad (9)$$

Thus the entries of the matrix and the right-hand side are obtained from local element integrals. Throughout the paper, we will use the grid size h to label the resulting approximations.

The Galerkin projection leads to a linear system of equations,

$$A_h \mathbf{x}_h = \mathbf{b}_h, \quad A_h = A_h^\dagger \succ 0. \quad (10)$$

In the quantum-algorithm analysis below, A_h , G_h , and $\mathbf{b}_h$ are understood in the discrete L^2 -normalized coordinate system, so that Euclidean vector norms represent the corresponding discrete L^2 norms. Equivalently, the usual nodal stiffness and load vectors are transformed by the finite element mass matrix. We keep the notation $A_h \mathbf{x}_h = \mathbf{b}_h$ for the normalized system. In these coordinates,

$$A_h = G_h^\dagger G_h, \quad \lambda_{\min}(A_h) = \Theta(1), \quad \lambda_{\max}(A_h) = \Theta(h^{-2}).$$

For second-order elliptic operators on quasi-uniform meshes, this gives $\kappa(A_h) = \Theta(h^{-2})$. Under the standard elliptic regularity assumption $u \in H^2(\Omega)$,

$$\|u - u_h\|_{L^2(\Omega)} \leq Ch^2 \|u\|_{H^2(\Omega)}. \quad (11)$$

$f \in L^2(\Omega)$ is a natural baseline assumption for obtaining the $\mathcal{O}(h^2)$ L^2 -rate [8, Sec. 5.7].

2.2 Divergence–gradient factorization

One key structural feature is that the stiffness matrix in (9) admits a Gram factorization

$$A_h = G_h^\dagger G_h. \quad (12)$$

For finite elements, this factorization can be seen directly from the element quadrature. Suppose the element integrals are evaluated using quadrature nodes x_{Kq} and weights w_{Kq} , where K indexes elements and q indexes quadrature points. Then G_h can be assembled from the gradient of the shape functions evaluated at x_{Kq} and scaled by w_{Kq} . In particular G_h can be regarded

as the weighted discrete gradient, and $G_h^\dagger$ is the corresponding discrete negative divergence. For conforming Dirichlet discretizations on quasi-uniform meshes, one has

$$\|G_h\| = \Theta(h^{-1}), \quad \sigma_{\min}(G_h) = \Theta(1). \quad (13)$$

We will consider this regime in this paper and highlight the scaling with the grid size h .

More generally, mimetic finite difference and compatible discretization methods are designed to preserve discrete gradient–divergence adjointness and integration-by-parts identities as well [9, 21], and they can also be considered. For highly structured finite difference or tensor-product discretizations, one may also exploit arithmetic structure directly in the block encoding, rather than relying only on generic sparse oracles; see, for example, explicit circuit constructions for structured sparse matrices and structured data-loading approaches [10, 37].

2.3 Sparse access and block encodings of G_h

We use the standard sparse-access model for matrices. The following lemma is a standard consequence of sparse block-encoding constructions and matrix arithmetic in the block-encoding/QSVT framework of Gilyén et al. [18].

Lemma 2.1 (Sparse-access block encoding). *Let $M \in \mathbb{C}^{N_1 \times N_2}$ be s -sparse by rows and columns. Assume access to oracles that, given a row or column index and an integer $\ell \in \{1, \dots, s\}$, return the location and value of the ℓ -th nonzero entry to precision ε_{val} . If $|M_{ij}| \leq M_{\max}$, then one can construct an (α, a, ε) -block encoding of M with*

$$\alpha = \mathcal{O}(sM_{\max}), \quad a = \mathcal{O}(\log(N_1 + N_2)),$$

using $\mathcal{O}(1)$ calls to the sparse-access oracles and additional gates $\text{polylog}((N_1 + N_2)/\varepsilon)$.

For finite differences and fixed-order finite elements, every row and column of G_h has only $\mathcal{O}(1)$ nonzeros (or $\mathcal{O}(d)$ for high-dimensional problems). After mass scaling, the nonzero entries of the discrete gradient are $\mathcal{O}(h^{-1})$, and (13) shows that this is also the correct operator-norm scale. Thus G_h admits a sparse-access block encoding with normalization

$$\alpha_G = \mathcal{O}(h^{-1}).$$

The use of G_h is also consistent with the general Gram-matrix constructions in the block-encoding and QSVT framework: Given a block encoding of G_h/α_G , one can obtain a block encoding of

$$G_h^\dagger G_h / \alpha_G^2$$

by standard block-encoding composition [18]. Thus the abstract construction of a Gram matrix from an encoded factor is already part of the quantum matrix-arithmetic toolkit. On the other hand, the ODE method developed below does not require a direct block encoding of the stiffness matrix $A_h = G_h^\dagger G_h$.

Corollary 2.2 (Block encoding of the first-order gradient operator). *Assume fixed spatial dimension, a shape-regular uniform mesh, fixed polynomial degree in the shape function, and bounded uniformly elliptic coefficients. Then G_h and $G_h^\dagger$ admit sparse-access block encodings with normalization*

$$\alpha_G = \mathcal{O}(h^{-1}).$$

Consequently, the Hermitian first-order matrix

$$H_G = \begin{bmatrix} 0 & -iG_h^\dagger \\ iG_h & 0 \end{bmatrix} = H_G^\dagger \quad (14)$$

admits a block encoding with normalization $\alpha_H = \mathcal{O}(h^{-1})$, using $\mathcal{O}(1)$ calls to block encodings of G_h and $G_h^\dagger$.

Remark 2.3. *Sparse access to G_h can be organized locally. Given a basis-function index, one enumerates the elements touching its support; given an element, one enumerates its local basis functions, quadrature nodes, and quadrature weights; and the value oracle computes the corresponding entry in (9). Thus sparse access is to the local element and quadrature data, not to a preassembled matrix. The same construction extends to higher-order finite elements when the solution and coefficient regularity justify them. The constants then depend on the polynomial degree p , but for fixed p the mesh dependence of the first-order factor remains $\mathcal{O}(h^{-1})$.*

2.4 Worst-case estimates based on spectral windows

Many QLSAs implicitly use spectral transformations. Let

$$A_h \mathbf{v}_j = \lambda_j \mathbf{v}_j, \quad \mathbf{b}_h = \sum_j b_j \mathbf{v}_j, \quad \mathbf{x}_* = A_h^{-1} \mathbf{b}_h = \sum_j \frac{b_j}{\lambda_j} \mathbf{v}_j.$$

In HHL-type algorithms [20], phase estimation must resolve eigenvalues throughout the promised spectral range, and the controlled reciprocal map $\lambda \mapsto 1/\lambda$ must be implemented uniformly on that range. In QSVT-based QLSA, the same issue appears as a polynomial approximation problem: one constructs a polynomial that approximates $1/\lambda$ on a normalized interval such as

$$\lambda \in [1/\kappa, 1].$$

The degree of this polynomial is governed by the worst-case endpoint $1/\kappa$. Thus the algorithm is designed to succeed uniformly over the promised interval, even if the input vector has very little weight in the most ill-conditioned modes [38].

It is useful to contrast this with classical solvers, e.g., Krylov or conjugate gradient, which can also be regarded as a polynomial approximation method: after k iterations,

$$\mathbf{x}_k \in \mathbf{x}_0 + \mathcal{K}_k(A_h, \mathbf{r}_0), \quad \mathcal{K}_k(A_h, \mathbf{r}_0) = \text{span}\{\mathbf{r}_0, A_h \mathbf{r}_0, \dots, A_h^{k-1} \mathbf{r}_0\},$$

where $\mathbf{r}_0 = \mathbf{b}_h - A_h \mathbf{x}_0$. These polynomials, however, are not fixed in advance by a worst-case approximation problem on the full spectral interval. They are generated by an orthogonalization process starting from the actual residual $\mathbf{r}_0$, and therefore depend on the spectral measure of A_h induced by the particular right-hand side and initial guess [35]. Just as importantly, conjugate gradient carries $\mathbf{r}_k$ throughout the computation and uses $\|\mathbf{r}_k\|$, as a built-in stopping test. Thus classical methods combine polynomial approximation with an adaptive residual-based stopping rule [35, 19].

For elliptic PDE inputs, the worst-case design may be conservative, as alluded to in the introduction. If f is smooth, then the coefficients b_j often decay rapidly with the frequency of the eigenmode, thus leading to a much smaller effective spectral window. This naturally leads to the interesting question of whether the residual error can be built into a quantum algorithm. The algebraic meaning of the residual is simple. Given an approximate solution $\mathbf{x}_h$, define

$$\mathbf{r} = \mathbf{b}_h - A_h \mathbf{x}_h.$$

If $\mathbf{x}_*$ and $\mathbf{e} = \mathbf{x}_* - \mathbf{x}_h$ denotes the actual solution and the error, respectively, then

$$A_h \mathbf{e}_h = \mathbf{r}.$$

Thus the residual is directly tied to the error equation and provides an a posteriori indicator of convergence.

3 ODE-based solvers and residual dynamics

We now turn the elliptic linear system

$$A_h \mathbf{x}_* = \mathbf{b}_h, \quad A_h = G_h^\dagger G_h,$$

into a stable first-order dynamical system. The construction is equation-based: instead of applying a QLSA directly to the stiffness matrix A_h , we use the divergence–gradient factor G_h to build a relaxation dynamics whose limiting solution block is the Galerkin solution.

3.1 Accumulator via ODE relaxation

The factorization $A_h = G_h^\dagger G_h$ says that the solution can be characterized by the first-order relations

$$\mathbf{q} = G_h \mathbf{x}, \quad G_h^\dagger \mathbf{q} = \mathbf{b}_h. \quad (15)$$

Here $\mathbf{q}$ is the discrete gradient, or flux-like, variable. If one evolves the pair $(\mathbf{x}, \mathbf{q})$ directly and then prepares the normalized joint state, the $\mathbf{q}$ -block may dominate the norm because $\mathbf{q}_* = G_h \mathbf{x}_*$ and $\|G_h\| = \mathcal{O}(h^{-1})$. For state preparation, it is preferable to organize the dynamics so that $\mathbf{x}(t)$ itself is the accumulator block.

Define the first-order residuals

$$\mathbf{r} := \mathbf{b}_h - G_h^\dagger \mathbf{q}, \quad \mathbf{s} := G_h \mathbf{x} - \mathbf{q}. \quad (16)$$

Here $\mathbf{r}$ can be interpreted as the conservation residual and $\mathbf{s}$ as the flux residual. We keep $\mathbf{q}$ implicit and evolve

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{r}, \\ \dot{\mathbf{r}} = -G_h^\dagger \mathbf{s}, \\ \dot{\mathbf{s}} = G_h \mathbf{r} - \mathbf{s}, \end{cases} \quad \mathbf{x}(0) = 0, \quad \mathbf{r}(0) = \mathbf{b}_h, \quad \mathbf{s}(0) = 0. \quad (17)$$

When needed, the flux variable is recovered from

$$\mathbf{q}(t) := G_h \mathbf{x}(t) - \mathbf{s}(t). \quad (18)$$

To write the dynamics in a compact form, we let

$$\mathbf{w}(t) = \begin{bmatrix} \mathbf{r}(t) \\ \mathbf{s}(t) \end{bmatrix}, \quad P_r = \begin{bmatrix} I & 0 \end{bmatrix}, \quad M_h = \begin{bmatrix} 0 & -G_h^\dagger \\ G_h & -I \end{bmatrix}.$$

Then

$$\dot{\mathbf{x}} = P_r \mathbf{w}, \quad \dot{\mathbf{w}} = M_h \mathbf{w}. \quad (19)$$

Equivalently, with

$$\mathbf{z}(t) = \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{w}(t) \end{bmatrix}, \quad L_h = \begin{bmatrix} 0 & P_r \\ 0 & M_h \end{bmatrix}, \quad (20)$$

the dynamics can be written as,

$$\dot{\mathbf{z}} = L_h \mathbf{z}. \quad (21)$$

The generator has the first-order mesh scale

$$\|L_h\| \leq 1 + \|M_h\| = \mathcal{O}(h^{-1}). \quad (22)$$

The unbounded mesh dependence enters only through G_h and $G_h^\dagger$.

Below we use the standard elliptic finite element setting: the space is conforming, the mesh is shape-regular and quasi-uniform, and homogeneous Dirichlet boundary conditions are imposed.

Theorem 3.1 (Exponential stability). *Let $(\mathbf{x}(t), \mathbf{w}(t))$ solve (19) with*

$$\mathbf{x}(0) = 0, \quad \mathbf{w}(0) = \begin{bmatrix} \mathbf{b}_h \\ 0 \end{bmatrix}.$$

Then M_h is uniformly exponentially stable: there exist constants $C_{\text{st}} \geq 1$ and $c_{\text{st}} > 0$, independent of h , such that

$$\|e^{tM_h}\| \leq C_{\text{st}} e^{-c_{\text{st}} t}, \quad t \geq 0. \quad (23)$$

Consequently,

$$\|\mathbf{w}(t)\| \leq C_{\text{st}} e^{-c_{\text{st}} t} \|\mathbf{w}(0)\|. \quad (24)$$

Moreover, $\mathbf{x}(t)$ converges to a limit $\mathbf{x}_$, and this limit solves*

$$G_h^\dagger G_h \mathbf{x}_* = \mathbf{b}_h.$$

In particular,

$$\|\mathbf{x}_* - \mathbf{x}(t)\| \leq C_{\text{tail}} C_{\text{st}} e^{-c_{\text{st}} t} \|\mathbf{w}(0)\|, \quad (25)$$

where

$$C_{\text{tail}} := \frac{C_{\text{st}}}{c_{\text{st}}}.$$

The proof is postponed to [Section A](#).

The theorem holds for any discrete vector $\mathbf{b}_h$. When $\mathbf{b}_h$ is obtained from a function f , the finite element estimate (11) describes the discretization error, while (25) describes the finite-time relaxation error.

[Eq. \(20\)](#) will be referred to as an accumulator, which on one hand encodes the solution of [Eq. \(10\)](#), and on the other hand, gives a useful output-block estimate. Define, for nonzero $\mathbf{x}_*$,

$$\Gamma_b := \frac{\|\mathbf{w}(0)\|}{\|\mathbf{x}_*\|} = \frac{\|\mathbf{b}_h\|}{\|\mathbf{x}_*\|}. \quad (26)$$

Thus choosing

$$T \geq c_{\text{st}}^{-1} \log \left(\frac{4C_{\text{st}}(1 + C_{\text{tail}})\Gamma_b}{\varepsilon} \right)$$

ensures

$$\|\mathbf{x}(T) - \mathbf{x}_*\| \leq \frac{\varepsilon}{4} \|\mathbf{x}_*\|, \quad \|\mathbf{w}(T)\| \leq \frac{\varepsilon}{4} \|\mathbf{x}_*\|.$$

Consequently,

$$\|\mathbf{x}(T)\| = (1 + \mathcal{O}(\varepsilon)) \|\mathbf{x}_*\|,$$

and the solution block $\mathbf{z}(t)$ has order-one weight in the joint state:

$$p_x(T) := \frac{\|\mathbf{x}(T)\|^2}{\|\mathbf{x}(T)\|^2 + \|\mathbf{w}(T)\|^2} = 1 - \mathcal{O}(\varepsilon^2). \quad (27)$$

Thus extracting the $\mathbf{x}$ -block from the accumulator state has constant overhead once the ODE has relaxed.

3.2 Quantum implementation and first-order simulation scale

The accumulator ODE (21) is the dynamical system implemented by the quantum ODE primitive. It contains the solution accumulator $\mathbf{x}(t)$ and the decaying residual block $\mathbf{w}(t)$. Since the ODE dynamics are non-unitary, the simulation complexity usually depends on the non-Hermitian structure of the generator [26, 2, 30]. For (21), the generator admits the following decomposition,

$$M_h = -iH_h^{(1)} - \Pi_s, \quad H_h^{(1)} = \begin{bmatrix} 0 & -iG_h^\dagger \\ iG_h & 0 \end{bmatrix}, \quad \Pi_s = \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix}.$$

In particular, the PDE-dependent Hermitian part has norm

$$\|H_h^{(1)}\| = \|G_h\| = \mathcal{O}(h^{-1}),$$

while the non-Hermitian damping has norm $\mathcal{O}(1)$. Equivalently, the full accumulator generator has the scale separation

$$L_h = -iH_{\text{big}} + K_{\text{big}},$$

where

$$H_{\text{big}} = \begin{bmatrix} 0 & 0 \\ 0 & H_h^{(1)} \end{bmatrix}, \quad K_{\text{big}} = \begin{bmatrix} 0 & P_r \\ 0 & -\Pi_s \end{bmatrix},$$

with

$$\|H_{\text{big}}\| = \mathcal{O}(h^{-1}), \quad \|K_{\text{big}}\| = \mathcal{O}(1).$$

The fact that the non-Hermitian part has much smaller scaling of $\mathcal{O}(1)$ leads to efficient simulation algorithms [26, 2, 30].

We now switch to quantum-state notation

$$|\mathbf{b}_h\rangle := \frac{\mathbf{b}_h}{\|\mathbf{b}_h\|}, \quad |\mathbf{x}_*\rangle := \frac{\mathbf{x}_*}{\|\mathbf{x}_*\|},$$

and invoke quantum ODE solvers.

Theorem 3.2 (QLSA via a first-order quantum ODE solver). *Assume sparse-access or block-encoding access to G_h as in Corollary 2.2, and assume that $\mathbf{b}_h$ is normalized and $|\mathbf{b}_h\rangle$ can be prepared with cost C_b . Consider the accumulator ODE (21) with initial state*

$$\mathbf{z}(0) = \begin{bmatrix} 0 \\ \mathbf{b}_h \\ 0 \end{bmatrix}.$$

Let T be chosen so that $|\mathbf{x}(T)\rangle$ is ε -close to $|\mathbf{x}_\rangle$ according to Eq. (25). Then an optimal linear-ODE primitive for nonunitary dynamics, such as the LCHS framework [2], can prepare an ε -accurate normalized solution state with leading cost*

$$\tilde{\mathcal{O}}\left(\frac{\|\mathbf{b}_h\|}{\|\mathbf{x}_*\|} \left[C_b + h^{-1} \text{polylog} \frac{1}{\varepsilon}\right]\right), \quad (28)$$

up to method-dependent logarithmic factors. If $\mathbf{b}_h$ is normalized before state preparation, the prefactor is equivalently the standard prefactor in QLSA

$$\frac{1}{\|A_h^{-1}|\mathbf{b}_h\rangle\|}.$$

Moreover, after relaxation the $\mathbf{x}$ -block has probability $p_x(T) = 1 - \mathcal{O}(\varepsilon^2)$ in the joint state. Thus extracting the solution block adds only constant overhead and does not introduce an additional h -dependent factor.

The proof is postponed to [Section B](#).

Remark 3.3. The prefactor $\|\mathbf{b}_h\|/\|\mathbf{x}_*\|$, or equivalently $1/\|A_h^{-1}|\mathbf{b}_h\rangle\|$ for normalized input, is the standard output-normalization factor for preparing a normalized state after nonunitary evolution. It is not caused by the first-order auxiliary variables. For elliptic problems, this prefactor has a well defined limit: as the ratio between the L^2 norms of f and u .

This dynamic approach makes another interesting connection between QLSA and quantum ODE algorithms: some ODE algorithms, by using history states, can be reduced to QLSA [\[6, 28\]](#). The current algorithm demonstrates that some QLS problems can be solved using quantum ODE solvers.

4 Residual measurements and dynamic stopping

Recall from the previous section that the accumulator residual-solution dynamics are

$$\mathbf{z}(t) = \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{w}(t) \end{bmatrix}, \quad \dot{\mathbf{z}} = L_h \mathbf{z}, \quad \mathbf{w} = (\mathbf{r}, \mathbf{s})^T.$$

Here $\mathbf{x}(t)$ is the solution accumulator and $\mathbf{w}(t)$ is the mixed residual block. The purpose of this section is to use the $\mathbf{w}$ -block as a measurable stopping signal. Instead of choosing the runtime only from a worst-case spectral estimate, we estimate the residual-block amplitude and use the stability estimate to decide when the current instance has entered the final exponentially decaying regime.

4.1 Residual probability from the joint state

We use bra-ket notation for the amplitude-encoded blocks of the joint vector. Thus $|\mathbf{x}(t)\rangle$ and $|\mathbf{w}(t)\rangle$ denote the computational-basis amplitude vectors associated with $\mathbf{x}(t)$ and $\mathbf{w}(t)$, with the overall normalization written explicitly. The dilated joint dynamics prepares, up to implementation error, a state of the form

$$|\psi(t)\rangle = \frac{|0\rangle |\mathbf{x}(t)\rangle + |1\rangle |\mathbf{w}(t)\rangle}{\sqrt{\|\mathbf{x}(t)\|^2 + \|\mathbf{w}(t)\|^2}}. \quad (29)$$

Let

$$\Pi_{\text{res}} = |1\rangle \langle 1| \otimes I$$

be the projector onto the residual block. Then

$$p_{\text{res}}(t) := \langle \psi(t) | \Pi_{\text{res}} | \psi(t) \rangle = \frac{\|\mathbf{w}(t)\|^2}{\|\mathbf{x}(t)\|^2 + \|\mathbf{w}(t)\|^2}. \quad (30)$$

The connection between this probability and convergence is the tail estimate from [Theorem 3.1](#):

$$\|\mathbf{x}_* - \mathbf{x}(t)\| \leq C_{\text{tail}} \|\mathbf{w}(t)\|, \quad C_{\text{tail}} := \frac{C_{\text{st}}}{c_{\text{st}}}. \quad (31)$$

Thus the residual block is not merely diagnostic; it controls the remaining distance to the steady state.

Lemma 4.1 (Residual probability controls the relative tail). *Assume $\|\mathbf{x}(t)\| > 0$. Fix $p \in (0, 1)$. If*

$$p_{\text{res}}(t) \leq p,$$

then

$$\frac{\|\mathbf{x}_* - \mathbf{x}(t)\|}{\|\mathbf{x}(t)\|} \leq C_{\text{tail}} \sqrt{\frac{p}{1-p}}.$$

Proof. From (30),

$$\frac{\|\mathbf{w}(t)\|^2}{\|\mathbf{x}(t)\|^2} = \frac{p_{\text{res}}(t)}{1 - p_{\text{res}}(t)}.$$

Combining this identity with (31) gives the claim. $\square$

4.2 Constant-threshold readout and deterministic coasting

A direct stopping rule based on the final tolerance would require

$$p_{\text{res}}(t) \lesssim \frac{\varepsilon^2}{C_{\text{tail}}^2 + \varepsilon^2},$$

which is an $\mathcal{O}(\varepsilon^2)$ -probability event. Estimating such a small probability by direct sampling would require $\mathcal{O}(\varepsilon^{-2})$ shots. We avoid this by using the exponential decay of the residual block.

Choose a fixed constant threshold $p_0 \in (0, 1)$, independent of ε , and define

$$\eta_0 := \sqrt{\frac{p_0}{1-p_0}}, \quad \beta_0 := C_{\text{tail}} \eta_0.$$

We choose p_0 so that

$$\beta_0 < 1.$$

Suppose that a checkpoint time t_k satisfies

$$p_{\text{res}}(t_k) \leq p_0.$$

Then

$$\|\mathbf{w}(t_k)\| \leq \eta_0 \|\mathbf{x}(t_k)\|$$

and, by Lemma 4.1,

$$\|\mathbf{x}_* - \mathbf{x}(t_k)\| \leq \beta_0 \|\mathbf{x}(t_k)\|.$$

Since $\beta_0 < 1$, the current accumulator norm is comparable to the solution norm:

$$\|\mathbf{x}(t_k)\| \leq \frac{1}{1 - \beta_0} \|\mathbf{x}_*\|.$$

Therefore,

$$\|\mathbf{w}(t_k)\| \leq \frac{\eta_0}{1 - \beta_0} \|\mathbf{x}_*\|.$$

Now continue the same ODE for an additional deterministic time Δt . Set

$$T_\star = t_k + \Delta t.$$

The residual decay gives

$$\|\mathbf{w}(T_\star)\| \leq C_{\text{st}} e^{-c_{\text{st}} \Delta t} \|\mathbf{w}(t_k)\|.$$

Combining this with the tail estimate gives

$$\|\mathbf{x}_* - \mathbf{x}(T_*)\| \leq \frac{C_{\text{tail}}C_{\text{st}}\eta_0}{1 - \beta_0} e^{-c_{\text{st}}\Delta t} \|\mathbf{x}_*\|.$$

Thus it is sufficient to choose

$$\Delta t \geq c_{\text{st}}^{-1} \log \left(\frac{C_{\text{tail}}C_{\text{st}}\eta_0}{(1 - \beta_0)\varepsilon} \right) \quad (32)$$

to ensure the relative error reaches the given precision

$$\|\mathbf{x}_* - \mathbf{x}(T_*)\| \leq \varepsilon \|\mathbf{x}_*\|.$$

This calculation suggests the stopping strategy used below: the checkpoint measurements only test for entry into a constant-residual regime, while the final accuracy is supplied by a deterministic coasting time Δt computed from the stability estimate. Hence the sampling problem is a constant-probability readout, rather than an ε^2 -probability readout.

Theorem 4.2 (Constant-threshold dynamic stopping). *Let $p_0 \in (0, 1)$ satisfy*

$$C_{\text{tail}} \sqrt{\frac{p_0}{1 - p_0}} < 1.$$

Suppose that t_{ent} is the first checkpoint time at which

$$p_{\text{res}}(t_{\text{ent}}) \leq p_0. \quad (33)$$

Let $T_ = t_k + \Delta t$, with Δt chosen as in (32). Then*

$$\|\mathbf{x}_* - \mathbf{x}(T_*)\| \leq \varepsilon \|\mathbf{x}_*\|.$$

Moreover, the final joint state

$$\begin{bmatrix} \mathbf{x}(T_*) \\ \mathbf{w}(T_*) \end{bmatrix}$$

has solution-block probability $1 - \mathcal{O}(\varepsilon^2)$.

4.3 Checkpoint sampling

Now we explain how to detect the check point that reaches the threshold [Eq. \(33\)](#).

Fix a checkpoint schedule

$$0 < t_0 < t_1 < \dots$$

For each candidate time t_k , a checkpoint circuit prepares a fresh copy of the initial state, evolves the dilated ODE to time t_k , measures the residual flag, and then discards the state. Thus one checkpoint circuit at time t_k produces one Bernoulli sample with success probability

$$p_k := p_{\text{res}}(t_k).$$

If N_{shot} independent checkpoint circuits are run at the same time t_k , then the outcome

$$X_k \sim \text{Binomial}(N_{\text{shot}}, p_k), \quad \hat{p}_k = \frac{X_k}{N_{\text{shot}}}.$$

The stopping-test stage examines the checkpoint times sequentially. At time t_k , it accepts that checkpoint if the estimated residual probability is below the constant threshold p_0 . A simple rule is

$$\hat{p}_k \leq p_0.$$

More conservatively, one may use a confidence buffer δ and accept only when

$$\hat{p}_k + \delta \leq p_0, \quad \delta = \sqrt{\frac{\log(2K/\nu)}{2N_{\text{shot}}}}, \quad (34)$$

where K is the number of checkpoint times tested and ν is an acceptable failure probability. The first index satisfying this buffered test is denoted

$$\hat{k}_{\text{ent}}, \quad \hat{t}_{\text{ent}} := t_{\hat{k}_{\text{ent}}}.$$

The final production time is then

$$T_{\star} = \hat{t}_{\text{ent}} + \Delta t,$$

where Δt is the deterministic coasting time from (32).

By Hoeffding's inequality and a union bound over the K checkpoint times, the buffered rule controls the probability of falsely accepting a checkpoint whose true residual probability is above p_0 by at most ν . For a fixed confidence buffer, this gives

$$N_{\text{shot}} = \mathcal{O}(\log(K/\nu)).$$

Thus the stopping-test sampling overhead is logarithmic in the number of checkpoints and the failure-probability parameter, and it is not polynomial in $1/\varepsilon$.

For a geometric checkpoint schedule $t_k = 2^k t_0$, the number of tested checkpoints is logarithmic in the accepted entry time t_{ent} . Since the final coasting time satisfies $\Delta t = \mathcal{O}(\log(1/\varepsilon))$, the total evolution time retains the same logarithmic precision dependence as the a priori stability bound, while the measured entry time remains instance dependent.

4.4 Algorithmic summary

The residual-based solver separates the stopping-test stage from the final production run. The stopping-test stage scans a sequence of checkpoint times $t_0, t_1, \dots$. At each candidate time t_k , it performs N_{shot} independent checkpoint runs. A single checkpoint run means: prepare the initial joint residual-solution state, evolve the dilated ODE once to time t_k , measure the residual flag, record one binary outcome, and discard the state. Thus the residual measurement is not used to continue the same quantum trajectory; it is only a Bernoulli sample for estimating $p_{\text{res}}(t_k)$.

Once the first accepted checkpoint time $\hat{t}_{\text{ent}}$ is selected, the final production run evolves a fresh initial state directly to

$$T_{\star} = \hat{t}_{\text{ent}} + \Delta t,$$

where Δt is the deterministic coasting time in (32). The production run is not interrupted by residual measurements.

We summarize the algorithm in [Algorithm 1](#). The checkpoint circuits are used only to detect when the current instance has entered a constant-residual regime. The final accuracy is then obtained by deterministic coasting under the proven exponential stability of the residual block. The final solution state is produced in a separate run.

[Figure 2](#) shows a possible workflow on hardware that supports mid-circuit measurement. The reset prepares the residual ancilla for the next checkpoint shot; it is not a refresh of an ongoing time evolution. Since each checkpoint run evolves directly from the initial state to t_k , the dilation only needs one time segment per shot.

Algorithm 1 Residual-based quantum elliptic solver with dynamic stopping

Require: Sparse-access or block-encoding access to G_h ; state preparation for $|\mathbf{b}_h\rangle$; target precision ε ; constant checkpoint threshold p_0 ; checkpoint times $t_0 < t_1 < t_2 < \dots$; number of checkpoint samples N_{shot} .

Ensure: A normalized solution state $|\tilde{\mathbf{x}}_h\rangle$ satisfying the stopping rule in [Theorem 4.2](#).

1: Construct the joint accumulator ODE

$$\dot{\mathbf{z}} = L_h \mathbf{z}, \quad \mathbf{z} = (\mathbf{x}, \mathbf{w})^T,$$

and its dilation to a Schrödinger equation.

2: **for** $k = 0, 1, 2, \dots$ **do**

3: Run N_{shot} independent checkpoint circuits: prepare the initial joint state, evolve once to time t_k , and measure the residual flag.

4: Let $\hat{p}_{\text{res}}(t_k)$ be the empirical frequency of the residual outcome.

5: **if** $\hat{p}_{\text{res}}(t_k) + \delta \leq p_0$, with δ as in [\(34\)](#) **then**

6: Set $t_{\text{ent}} \leftarrow t_k$ and exit the loop.

7: **end if**

8: **end for**

9: Set $T_\star \leftarrow t_{\text{ent}} + \Delta t$, with Δt chosen by [\(32\)](#).

10: Prepare a fresh initial state and evolve the dilated joint system once to time $T_\star$, without residual measurements during the evolution.

11: Output the solution block, namely the $\mathbf{x}$ -subregister of $\mathbf{z}(T_\star)$, with the appropriate normalization and postselection factors.

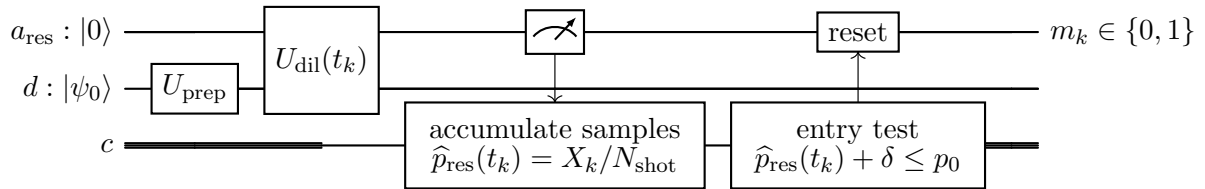


Figure 2: Checkpoint circuit for residual-based stopping. Each checkpoint run prepares the initial state, evolves the dilated joint accumulator system once to time t_k , and measures the residual flag a_{res} . The outcome $m_k \in \{0, 1\}$ is one Bernoulli sample. Repeating the checkpoint circuit N_{shot} times gives the estimate $\hat{p}_{\text{res}}(t_k) = X_k / N_{\text{shot}}$, which is compared with the constant threshold p_0 . If the entry test accepts, a separate production run evolves directly to $T_\star = t_k + \Delta t$, without residual measurements, and outputs the solution block.

5 Numerical experiments

We illustrate the residual-based stopping mechanism on a standard finite element discretization of the Poisson equation on the unit square ($a(x) = I$),

$$-\Delta u = f \quad \text{in } \Omega = (0, 1)^2, \quad u = 0 \quad \text{on } \partial\Omega. \quad (35)$$

The domain is partitioned into $n \times n$ square cells, and each square is split along the same diagonal to obtain a conforming triangular mesh. We use continuous piecewise-linear, or P_1 , finite elements, with homogeneous Dirichlet boundary conditions imposed by eliminating the boundary degrees of freedom. In the experiments below, $n = 16$, so $h = 1/16$. The mesh contains $16^2 = 256$ square cells and 512 triangular elements. After imposing the boundary condition, there are $(16 - 1)^2 = 225$ interior degrees of freedom. The discrete gradient has two components on each triangle, giving $2 \cdot 512 = 1024$ gradient degrees of freedom.

The test problem is chosen by the method of manufactured solutions. We take

$$u_{\text{ex}}(x, y) = \sin(\pi x) \sin(\pi y) + \frac{1}{2} \sin(3\pi x) \sin(2\pi y) + \frac{1}{4} \sin(5\pi x) \sin(4\pi y), \quad (36)$$

and define $f = -\Delta u_{\text{ex}}$. Thus

$$f(x, y) = 2\pi^2 \sin(\pi x) \sin(\pi y) + \frac{13\pi^2}{2} \sin(3\pi x) \sin(2\pi y) + \frac{41\pi^2}{4} \sin(5\pi x) \sin(4\pi y). \quad (37)$$

This gives a smooth right-hand side with several spectral components, while still allowing comparison with a known exact solution. The mesh and the corresponding finite element solution are shown in Fig. 3.

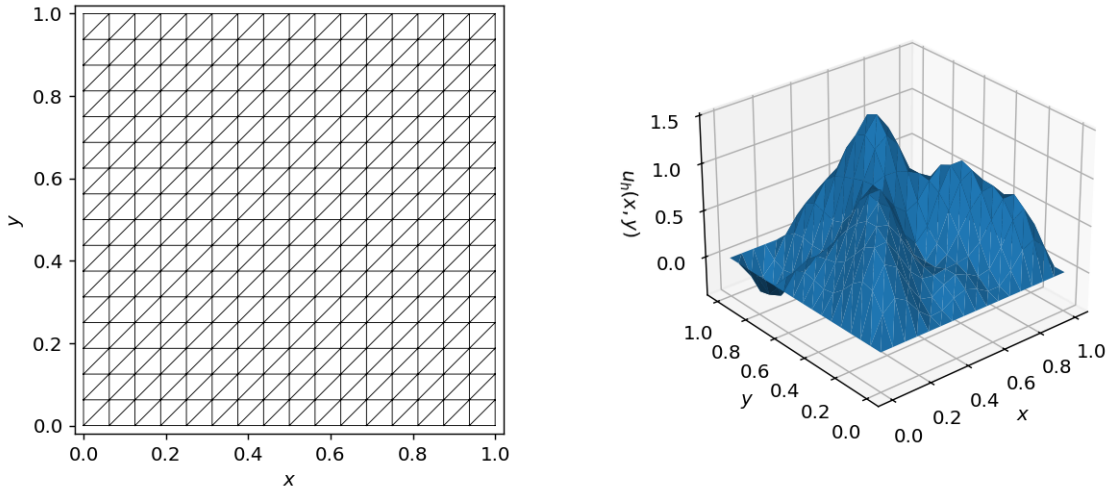


Figure 3: Finite element setup for the model Poisson problem. Left: the uniform triangular mesh obtained by splitting each square cell into two triangles. Right: the P_1 finite element solution u_h for the manufactured solution in (36). The factorization $A_h = G_h^\dagger G_h$ is assembled from the elementwise gradients of the P_1 basis functions on this mesh.

We next evolve the residual-augmented first-order system in Eq. (17). The flux variable is recovered as

$$\mathbf{q}(t) = G_h \mathbf{x}(t) - \mathbf{s}(t).$$

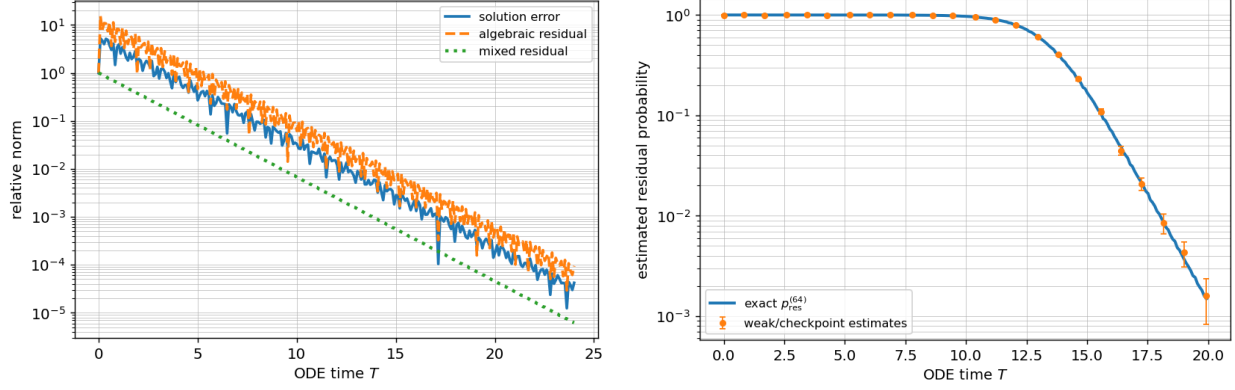


Figure 4: Residual dynamics and checkpoint readout. Left: relative solution error, relative algebraic residual $\|\mathbf{r}_A(t)\|/\|\mathbf{b}_h\|$, mixed residual $\|\mathbf{w}(t)\|/\|\mathbf{w}(0)\|$, and accumulator residual probability $p_{\text{res}}(t)$, computed from $(\mathbf{x}(t), \mathbf{w}(t))$. Right: checkpoint estimates of $p_{\text{res}}(t)$ from Bernoulli samples of the residual flag.

Thus

$$\mathbf{r}(t) = \mathbf{b}_h - G_h^T \mathbf{q}(t), \quad \mathbf{s}(t) = G_h \mathbf{x}(t) - \mathbf{q}(t),$$

so $(\mathbf{r}, \mathbf{s})$ is the mixed first-order residual carried by the ODE. The algebraic residual is

$$\mathbf{r}_A(t) := \mathbf{b}_h - A_h \mathbf{x}(t) = \mathbf{r}(t) - G_h^\dagger \mathbf{s}(t).$$

For the residual register, we use $\mathbf{w} = (\mathbf{r}, \mathbf{s})$, and measure the probability

$$p_{\text{res}}(t) = \frac{\|\mathbf{w}(t)\|^2}{\|\mathbf{x}(t)\|^2 + \|\mathbf{w}(t)\|^2}. \quad (38)$$

Figure 4 shows the main numerical mechanism. The algebraic residual and the mixed residual decay together with the solution error, up to the oscillations expected from the first-order damped dynamics. The residual-register probability $p_{\text{res}}(t)$ follows the same trend. Thus measuring the residual register gives an observable proxy for convergence, without reconstructing $\mathbf{x}(t)$ or evaluating $\mathbf{b}_h - A_h \mathbf{x}(t)$ classically. The right panel shows that this probability can be estimated directly from repeated checkpoint measurements.

6 Summary and Discussions

This paper develops a residual-based dynamic stopping mechanism for quantum linear-system algorithms, with elliptic PDEs as an important motivating application. The central idea is to make the residual a co-evolving part of the quantum dynamics rather than a quantity reconstructed after the solution state has been prepared. For elliptic discretizations with $A_h = G_h^\dagger G_h$, the first-order factor G_h gives the mesh-dependent scale $\|G_h\| = \mathcal{O}(h^{-1})$, while the residual variables are driven by the same stable first-order dynamics. After dilation to a Schrödinger evolution, the residual block can be measured in checkpoint runs, producing an instance-dependent stopping signal. Thus the contribution is not a new worst-case QLSA condition-number improvement, but a mechanism for making QLSA-based PDE solvers behave more like adaptive algorithms: run the dynamics, monitor a residual, and stop when the current instance has converged. Although the construction is formulated for elliptic PDEs, the residual-based stopping idea can also be extended to general

stable linear systems $A\mathbf{x} = \mathbf{b}$, by using an ODE relaxation in which the algebraic residual evolves under the same stable dynamics. The elliptic setting is special because the factorization $A_h = G_h^\dagger G_h$ exposes the first-order scale $\|G_h\| = \mathcal{O}(h^{-1})$, which a generic stable system need not provide.

Several implementation questions are interesting. Since the residual register is a stopping mechanism rather than a replacement for existing QLSA primitives, a complete resource estimate should separate the cost of the underlying nonunitary ODE solver, the output-normalization factor, the checkpoint sampling overhead, and the final solution readout. The accumulator formulation is useful in this respect: after relaxation the solution block has order-one weight, so extracting $\mathbf{x}(T)$ does not introduce an additional mesh-dependent block-selection penalty. At the same time, estimating absolute quantities may still require estimating $\|\mathbf{x}_*\|$, which can be obtained from the output-normalization or postselection amplitude of the ODE primitive. It may also be useful to combine the residual flag with the dilation ancillas for ODE algorithms [26, 5, 30], so that checkpoint shots estimate both the conditional residual probability and the simulation success probability. Finally, the constants in the stability estimate enter only when converting a measured residual signal into a rigorous error bound; conservative constants give a valid but possibly cautious coasting time, while the measured entry time into the small-residual regime remains instance dependent.

For many PDE applications, the desired output is not the whole solution state but a quantity of interest, such as an energy, boundary flux, averaged field value, or response functional. This suggests developing observable-specific stopping rules, possibly using dual-weighted or adjoint residuals. Another natural direction is nonsymmetric steady-state problems such as advection–diffusion, where the diffusion part still has a coercive $G_h^\dagger G_h$ structure while the transport term acts as a lower-order or skew perturbation. Understanding when such systems admit stable first-order dilations with measurable residual registers would make residual-based dynamic stopping applicable to a broader class of quantum PDE algorithms.

Acknowledgments

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A Proof of the ODE stability

Proof of Theorem 3.1. The residual block satisfies

$$\dot{\mathbf{w}} = M_h \mathbf{w}, \quad \mathbf{w} = \begin{bmatrix} \mathbf{r} \\ \mathbf{s} \end{bmatrix}, \quad M_h = \begin{bmatrix} 0 & -G_h^\dagger \\ G_h & -I \end{bmatrix}.$$

We prove the stability estimate by a Lyapunov functional argument. Let

$$A_h = G_h^\dagger G_h, \quad K_h := A_h^{-1} G_h^\dagger, \quad P_h := G_h A_h^{-1} G_h^\dagger.$$

The discrete Poincaré inequality gives

$$\|G_h \mathbf{v}\| \geq \gamma_0 \|\mathbf{v}\|,$$

with $\gamma_0 > 0$ independent of h . Hence A_h is positive definite. Moreover, P_h is the orthogonal projector onto $\text{Range}(G_h)$, and

$$K_h G_h = I.$$

Also,

$$\|K_h \mathbf{s}\| \leq \gamma_0^{-1} \|P_h \mathbf{s}\| \leq \gamma_0^{-1} \|\mathbf{s}\|.$$

Choose a constant $\eta > 0$, depending only on γ_0 , such that

$$0 < \eta < \gamma_0, \quad 2(1 - \eta) - \frac{\eta}{\gamma_0^2} > 0.$$

Define

$$\mathcal{E}(\mathbf{r}, \mathbf{s}) = \|\mathbf{r}\|^2 + \|\mathbf{s}\|^2 - 2\eta \text{Re}\langle \mathbf{r}, K_h \mathbf{s} \rangle.$$

Since

$$|\langle \mathbf{r}, K_h \mathbf{s} \rangle| \leq \gamma_0^{-1} \|\mathbf{r}\| \|\mathbf{s}\|,$$

the functional $\mathcal{E}$ is uniformly equivalent to $\|\mathbf{r}\|^2 + \|\mathbf{s}\|^2$:

$$\left(1 - \frac{\eta}{\gamma_0}\right) (\|\mathbf{r}\|^2 + \|\mathbf{s}\|^2) \leq \mathcal{E}(\mathbf{r}, \mathbf{s}) \leq \left(1 + \frac{\eta}{\gamma_0}\right) (\|\mathbf{r}\|^2 + \|\mathbf{s}\|^2).$$

We now use $\mathcal{E}$ as the Lyapunov functional and differentiate it along the dynamics

$$\dot{\mathbf{r}} = -G_h^\dagger \mathbf{s}, \quad \dot{\mathbf{s}} = G_h \mathbf{r} - \mathbf{s}.$$

First,

$$\frac{d}{dt} (\|\mathbf{r}\|^2 + \|\mathbf{s}\|^2) = -2\|\mathbf{s}\|^2.$$

Let

$$C(t) := \operatorname{Re}\langle \mathbf{r}(t), K_h \mathbf{s}(t) \rangle.$$

Using $K_h G_h = I$, we obtain

$$\begin{aligned} \dot{C}(t) &= \operatorname{Re}\langle \dot{\mathbf{r}}, K_h \mathbf{s} \rangle + \operatorname{Re}\langle \mathbf{r}, K_h \dot{\mathbf{s}} \rangle \\ &= -\operatorname{Re}\langle G_h^\dagger \mathbf{s}, A_h^{-1} G_h^\dagger \mathbf{s} \rangle + \operatorname{Re}\langle \mathbf{r}, K_h (G_h \mathbf{r} - \mathbf{s}) \rangle \\ &= -\|P_h \mathbf{s}\|^2 + \|\mathbf{r}\|^2 - C(t). \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{d}{dt} \mathcal{E} &= -2\|\mathbf{s}\|^2 - 2\eta \dot{C}(t) \\ &= -2\eta \|\mathbf{r}\|^2 - 2\|\mathbf{s}\|^2 + 2\eta \|P_h \mathbf{s}\|^2 + 2\eta C(t). \end{aligned}$$

Decompose

$$\mathbf{s} = P_h \mathbf{s} + (I - P_h) \mathbf{s}.$$

Using

$$2|C(t)| \leq \|\mathbf{r}\|^2 + \gamma_0^{-2} \|P_h \mathbf{s}\|^2,$$

we get

$$\frac{d}{dt} \mathcal{E} \leq -\eta \|\mathbf{r}\|^2 - \left(2(1 - \eta) - \frac{\eta}{\gamma_0^2} \right) \|P_h \mathbf{s}\|^2 - 2\|(I - P_h) \mathbf{s}\|^2.$$

By the choice of η , the right-hand side is bounded above by

$$-c_0 (\|\mathbf{r}\|^2 + \|\mathbf{s}\|^2)$$

for some $c_0 > 0$ independent of h . Since $\mathcal{E}$ is uniformly equivalent to $\|\mathbf{r}\|^2 + \|\mathbf{s}\|^2$, there exist constants $C_{\text{st}} \geq 1$ and $c_{\text{st}} > 0$, independent of h , such that

$$\|e^{tM_h}\| \leq C_{\text{st}} e^{-c_{\text{st}} t}, \quad t \geq 0.$$

This proves the stability estimate and hence

$$\|\mathbf{w}(t)\| \leq C_{\text{st}} e^{-c_{\text{st}} t} \|\mathbf{w}(0)\|.$$

Since $\mathbf{w}(t)$ is integrable in time, the accumulator

$$\mathbf{x}_* := \mathbf{x}(0) + \int_0^\infty P_r \mathbf{w}(\tau) d\tau$$

exists. Let $\mathbf{q}(t)$ be defined by (18). The propagated identities give

$$\mathbf{r}(t) = \mathbf{b}_h - G_h^\dagger \mathbf{q}(t), \quad \mathbf{s}(t) = G_h \mathbf{x}(t) - \mathbf{q}(t).$$

Taking $t \rightarrow \infty$ and using $\mathbf{r}(t), \mathbf{s}(t) \rightarrow 0$ gives

$$\mathbf{b}_h - G_h^\dagger \mathbf{q}_* = 0, \quad G_h \mathbf{x}_* - \mathbf{q}_* = 0.$$

Therefore

$$G_h^\dagger G_h \mathbf{x}_* = \mathbf{b}_h.$$

For the tail bound, use the semigroup property:

$$\mathbf{x}_* - \mathbf{x}(t) = \int_t^\infty P_r \mathbf{w}(\tau) d\tau = P_r \int_0^\infty e^{sM_h} \mathbf{w}(t) ds.$$

Since $\|P_r\| = 1$, the stability estimate gives

$$\|\mathbf{x}_* - \mathbf{x}(t)\| \leq \int_0^\infty C_{\text{st}} e^{-c_{\text{st}} s} ds \|\mathbf{w}(t)\| = \frac{C_{\text{st}}}{c_{\text{st}}} \|\mathbf{w}(t)\|.$$

□

B Proof of the ODE-based QLSA complexity

Proof of Theorem 3.2. By Corollary 2.2, the accumulator generator

$$L_h = \begin{bmatrix} 0 & P_r \\ 0 & M_h \end{bmatrix}, \quad M_h = \begin{bmatrix} 0 & -G_h^\dagger \\ G_h & -I \end{bmatrix},$$

has a block encoding with normalization

$$\alpha_L = \mathcal{O}(h^{-1}),$$

because it consists only of G_h , $G_h^\dagger$, identity blocks, the bounded projection P_r , and bounded damping/projector blocks. By Theorem 3.1, choosing

$$T = \tilde{\mathcal{O}}\left(\log \frac{1}{\varepsilon}\right)$$

makes the finite-time relaxation error $\mathcal{O}(\varepsilon)$, with logarithmic dependence on h and on $\|\mathbf{b}_h\|/\|\mathbf{x}_*\|$ absorbed in $\tilde{\mathcal{O}}$. Hence

$$\alpha_L T = \tilde{\mathcal{O}}\left(h^{-1} \log \frac{1}{\varepsilon}\right).$$

We next account for the output normalization. The accumulator ODE is homogeneous:

$$\mathbf{z}(T) = e^{TL_h} \mathbf{z}(0), \quad \mathbf{z}(0) = \begin{bmatrix} 0 \\ \mathbf{b}_h \\ 0 \end{bmatrix}.$$

For LCHS-type algorithms, the state-preparation overhead is governed by the output-norm ratio of the nonunitary evolution. In the homogeneous time-independent case, this is the factor

$$\Gamma_{\text{out}}(T) := \frac{\|\mathbf{z}(0)\|}{\|\mathbf{z}(T)\|}.$$

This is the homogeneous specialization of the standard LCHS output-normalization factor; compare the homogeneous-case corollaries of [2], where the state-preparation cost is optimal in this ratio.

In the present accumulator system,

$$\|\mathbf{z}(0)\| = \|\mathbf{b}_h\|, \quad \|\mathbf{z}(T)\|^2 = \|\mathbf{x}(T)\|^2 + \|\mathbf{w}(T)\|^2.$$

By the choice of T ,

$$\|\mathbf{x}(T) - \mathbf{x}_*\| = \mathcal{O}(\varepsilon) \|\mathbf{x}_*\|, \quad \|\mathbf{w}(T)\| = \mathcal{O}(\varepsilon) \|\mathbf{x}_*\|.$$

Therefore

$$\|\mathbf{z}(T)\| = (1 + \mathcal{O}(\varepsilon))\|\mathbf{x}_*\|,$$

and hence

$$\Gamma_{\text{out}}(T) = (1 + \mathcal{O}(\varepsilon)) \frac{\|\mathbf{b}_h\|}{\|\mathbf{x}_*\|}.$$

If the right-hand side is normalized before state preparation, then

$$\hat{\mathbf{x}}_* = A_h^{-1} |\mathbf{b}_h\rangle = \frac{\mathbf{x}_*}{\|\mathbf{b}_h\|},$$

so the same factor is

$$\Gamma_{\text{out}}(T) = (1 + \mathcal{O}(\varepsilon)) \frac{1}{\|A_h^{-1} |\mathbf{b}_h\rangle\|}.$$

Combining the LCHS query dependence on $\alpha_L T$, the input-state preparation cost C_b , and the output-normalization factor $\Gamma_{\text{out}}(T)$ gives

$$\tilde{\mathcal{O}}\left(\frac{\|\mathbf{b}_h\|}{\|\mathbf{x}_*\|} \left[C_b + h^{-1} \text{polylog} \frac{1}{\varepsilon}\right]\right).$$

It remains to check the cost of extracting the solution block. Since

$$\mathbf{z}(T) = \begin{bmatrix} \mathbf{x}(T) \\ \mathbf{w}(T) \end{bmatrix},$$

the probability of selecting the $\mathbf{x}$ -block is

$$p_x(T) = \frac{\|\mathbf{x}(T)\|^2}{\|\mathbf{x}(T)\|^2 + \|\mathbf{w}(T)\|^2}.$$

Using the bounds above,

$$\|\mathbf{x}(T)\| = (1 + \mathcal{O}(\varepsilon))\|\mathbf{x}_*\|, \quad \|\mathbf{w}(T)\| = \mathcal{O}(\varepsilon)\|\mathbf{x}_*\|,$$

so

$$p_x(T) = 1 - \mathcal{O}(\varepsilon^2).$$

Thus selecting the $\mathbf{x}$ -block succeeds with order-one probability and does not change the leading mesh dependence. Finally, since

$$\|\mathbf{x}(T) - \mathbf{x}_*\| = \mathcal{O}(\varepsilon)\|\mathbf{x}_*\|,$$

the normalized states $|\mathbf{x}(T)\rangle$ and $|\mathbf{x}_*\rangle$ are $\mathcal{O}(\varepsilon)$ -close. □